Programming Assignment Report

Design and Analysis of Algorithms:

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Problem Name: Beautiful Days at the Movies

Problem Statement:

Lily likes to play games with integers. She has created a new game where she determines the difference between a number and its reverse. For instance, given the number **12**, its reverse is **21**. Their difference is **9**. The number **120** reversed is **21**, and their difference is **99**.

She decides to apply her game to decision making. She will look at a numbered range of days and will only go to a movie on a *beautiful day*.
Given a range of numbered days**, [i...j]** and a number **k**, determine the number of days in the range that are *beautiful*. Beautiful numbers are defined as numbers where **|i -reverse(i)|** is evenly divisible by **k.** If a day's value is a beautiful number, it is a beautiful day. Return the number of beautiful days in the range.

Function Description

Complete the *beautiful Days* function in the editor below.

beautiful Days has the following parameter(s):

-> int i: the starting day number

-> ***int j:*** the ending day number

-> int k: the divisor

Problem Hacker Rank Link: https://www.hackerrank.com/challenges/beautiful-days-atthe-movies/problem

Github Submission Link: https://github.com/bhanuprakashkotha947/DAA-Assignment/tree/main

Solution Steps

1. Read Input

- Read three integers **i, j, and k**.
- These represent:
  - **i** → starting day
  - **j** → ending day
  - **k** → divisor

2. Initialize Counter

> Create a variable **count**.
> This will store the number of beautiful days.

3. Iterate Through Days

> Loop through each day from **i to j**.

for (day = i → j)

4. Reverse the Number

For each day:

>> Extract digits using modulus %
>>Build the reverse number.

Example: num = day
reverse = reverse * 10 + num % 10
num = num / 10

5. Calculate Absolute Difference

Compute: diff = |day - reverse|

6. Check Divisibility

Check if:

diff % k == 0

If true → the day is **beautiful**.

7. Increment Counter

> If the condition is satisfied:
count++

8. Return Result

> After checking all days:
> Return the total number of beautiful days.

Complete JAVA Code:

```
import java.io.*;

import java.util.*;

class Result {

   public static int beautifulDays(int i, int j, int k) {

      int count = 0;

      for (int day = i; day <= j; day++) {

         int num = day;

         int reverse = 0;

         // reverse the number

         while (num > 0) {

            reverse = reverse * 10 + num % 10;

            num /= 10;

         }

         int diff = Math.abs(day - reverse);
```

```
            if (diff % k == 0) {
                count++;
            }
        }
        return count;
    }
}
public class Solution {
    public static void main(String[] args) throws IOException {
        BufferedReader bufferedReader = new BufferedReader(new
InputStreamReader(System.in));
        String[] input = bufferedReader.readLine().split(" ");
        int i = Integer.parseInt(input[0]);
        int j = Integer.parseInt(input[1]);
        int k = Integer.parseInt(input[2]);
        int result = Result.beautifulDays(i, j, k);
        System.out.println(result);
        bufferedReader.close();
    }
}
```

Sample Input:

20 23 6

Sample Output:

2

Observation / Solution :

Key Idea

The main concept is to compare a number with its reverse and determine whether their difference is divisible by k.

Approach

1. Iterate through all numbers in the range i to j.
2. Reverse each number.
3. Calculate the absolute difference.
4. Check if the difference is divisible by k.
5. Count such numbers.

Complexity:

Time Complexity:

$$O(n \times d)$$

Where:

- **n** = number of days
- **d** = number of digits

Since digits are small, this is effectively:

$$O(n)$$

Space Complexity:

$$O(1)$$

Only a few variables are used.